B. Battling with Numbers

time limit per test: 1 second
memory limit per test: 256 megabytes

On the trip to campus during the mid semester exam period, Chaneka thinks of two positive integers X and Y. Since the two integers can be very big, both are represented using their prime factorisations, such that:

- $X = A_1^{B_1} \times A_2^{B_2} \times \ldots \times A_N^{B_N}$ (each A_i is prime, each B_i is positive, and $A_1 < A_2 < \ldots < A_N$)
- $Y = C_1^{D_1} \times C_2^{D_2} \times \ldots \times C_M^{D_M}$ (each C_j is prime, each D_j is positive, and $C_1 < C_2 < \ldots < C_M$)

Chaneka ponders about these two integers for too long throughout the trip, so Chaneka's friend commands her "Gece, deh!" (move fast) in order to not be late for the exam.

Because of that command, Chaneka comes up with a problem, how many pairs of positive integers p and q such that $\text{LCM}(p, q) = X$ and $\text{GCD}(p, q) = Y$. Since the answer can be very big, output the answer modulo $998\,244\,353$.

Notes:

- $\text{LCM}(p, q)$ is the smallest positive integer that is simultaneously divisible by p and q.
- $\text{GCD}(p, q)$ is the biggest positive integer that simultaneously divides p and q.

Input

The first line contains a single integer N ($1 \leq N \leq 10^5$) — the number of distinct primes in the prime factorisation of X.

The second line contains N integers $A_1, A_2, A_3, \ldots, A_N$ ($2 \leq A_1 < A_2 < \ldots < A_N \leq 2 \cdot 10^6$; each A_i is prime) — the primes in the prime factorisation of X.

The third line contains N integers $B_1, B_2, B_3, \ldots, B_N$ ($1 \leq B_i \leq 10^5$) — the exponents in the prime factorisation of X.

The fourth line contains a single integer M ($1 \leq M \leq 10^5$) — the number of distinct primes in the prime factorisation of Y.

The fifth line contains M integers $C_1, C_2, C_3, \ldots, C_M$ ($2 \leq C_1 < C_2 < \ldots < C_M \leq 2 \cdot 10^6$; each C_j is prime) — the primes in the prime factorisation of Y.

The sixth line contains M integers $D_1, D_2, D_3, \ldots, D_M$ ($1 \leq D_j \leq 10^5$) — the exponents in the prime factorisation of Y.

Output

An integer representing the number of pairs of positive integers p and q such that $\text{LCM}(p, q) = X$ and $\text{GCD}(p, q) = Y$, modulo $998\,244\,353$.

Examples

input Copy

```
4
2 3 5 7
2 1 1 2
2
3 7
1 1
```

output Copy

```
8
```

input Copy

```
2
1299721 1999993
100000 265
2
1299721 1999993
100000 265
```

output Copy

```
1
```

input Copy

```
2
2 5
1 1
2
2 3
1 1
```

output Copy

```
0
```

COMPFEST 15 - Preliminary Online Mirror (Unrated, ICPC Rules, Teams Preferred)
Finished
Practice

→ Virtual participation
Virtual contest is a way to take part in past contest, as close as possible to participation on time. It is supported only ICPC mode for virtual contests. If you've seen these problems, a virtual contest is not for you - solve these problems in the archive. If you just want to solve some problem from a contest, a virtual contest is not for you - solve this problem in the archive. Never use someone else's code, read the tutorials or communicate with other person during a virtual contest.
Start virtual contest

→ Clone Contest to Mashup
You can clone this contest to a mashup.
Clone Contest

→ Submit?
Language: GNU G++23 14.2 (64 bit, msys
Choose file: Choose File No file chosen
Submit

→ Last submissions

Submission	Time	Verdict
348886731	Nov/14/2025 15:23	Happy New Year!
348886630	Nov/14/2025 15:22	Delivered to wrong address on flight 15 WA15

→ Problem tags
combinatorics math number theory *1400
No tag edit access

→ Contest materials
- Announcement (en)
- Tutorial (en)

Note

In the first example, the integers are as follows:

- $X = 2^2 \times 3^1 \times 5^1 \times 7^2 = 2940$
- $Y = 3^1 \times 7^1 = 21$

The following are all possible pairs of p and q:

- $p = 21, q = 2940$
- $p = 84, q = 735$
- $p = 105, q = 588$
- $p = 147, q = 420$
- $p = 420, q = 147$
- $p = 588, q = 105$
- $p = 735, q = 84$
- $p = 2940, q = 21$

In the third example, the integers are as follows:

- $X = 2^1 \times 5^1 = 10$
- $Y = 2^1 \times 3^1 = 6$

There is no pair p and q that simultaneously satisfies $\mathrm{LCM}(p, q) = 10$ and $\mathrm{GCD}(p, q) = 6$.